

Computer Networks

Data Link Control Protocols (Medium Access Control Protocols) (Random Access Protocols: ALOHA)

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MAC protocols: taxonomy

Three broad classes:

- **Channel partitioning**

- Divide channel into smaller “pieces” (time slots, frequency, code)
- Allocate piece to node for exclusive use

- **Taking turns”**

- Nodes take turns, but nodes with more to send can take longer turns

- **Random access**

- Use randomization for handling collisions
 - “Recover” from collisions
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Random access protocols

- When node has packet to send
 - Transmit at full channel data rate R
 - No a priori coordination among nodes
 - Two or more transmitting nodes: “collision”
 - **Random access protocol** specifies:
 - How to detect collisions
 - How to recover from collisions (e.g., via delayed retransmissions)
 - Examples of random access MAC protocols:
 - ALOHA, slotted ALOHA
 - CSMA, CSMA/CD, CSMA/CA
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Slotted ALOHA

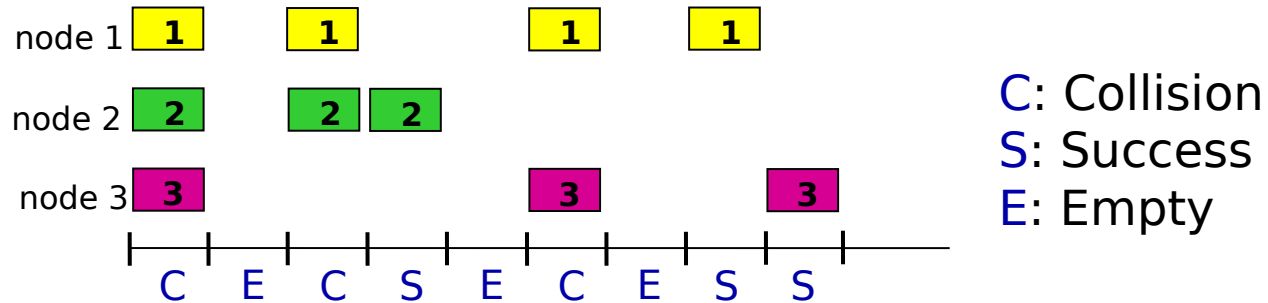
Assumptions:

- All frames same size
- Time divided into equal size slots (time to transmit 1 frame)
- Nodes start to transmit only slot beginning
- Nodes are synchronized
- If 2 or more nodes transmit in slot, all nodes detect collision

Operation:

- When node obtains fresh frame, transmits in next slot
 - **If no collision:** node can send new frame in next slot
 - **If collision:** node retransmits frame in each subsequent slot with probability **p** until success
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Slotted ALOHA



Pros:

- Single active node can continuously transmit at full rate of channel
- Highly decentralized: only slots in nodes need to be in sync
- Simple

Cons:

- Collisions, wasting slots
- Idle slots
- Clock synchronization

Slotted ALOHA: efficiency

Efficiency: Long-run fraction of successful slots
(many nodes, all with many frames to send)

- Suppose: N nodes with many frames to send, each transmits in slot with probability p
 - Prob that given node has success in a slot = $p(1-p)^{N-1}$
 - Prob that any node has a success = $Np(1-p)^{N-1}$
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Slotted ALOHA: efficiency

- Max efficiency: find p^* that maximizes $Np(1-p)^{N-1}$

- Max efficiency = $1/e = .37$
 - At best: channel used for useful transmissions 37% of time!
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Pure ALOHA efficiency

$$\begin{aligned} P(\text{success by given node}) &= P(\text{node transmits}) \\ &\quad \times P(\text{no other node transmits in} \\ &\quad [t_0-1, t_0]) \\ &\quad \times P(\text{no other node transmits in} \\ &\quad [t_0, t_0+1]) \\ &= p \cdot (1-p)^{N-1} \cdot (1-p)^{N-1} \\ &= p \cdot (1-p)^{2(N-1)} \end{aligned}$$

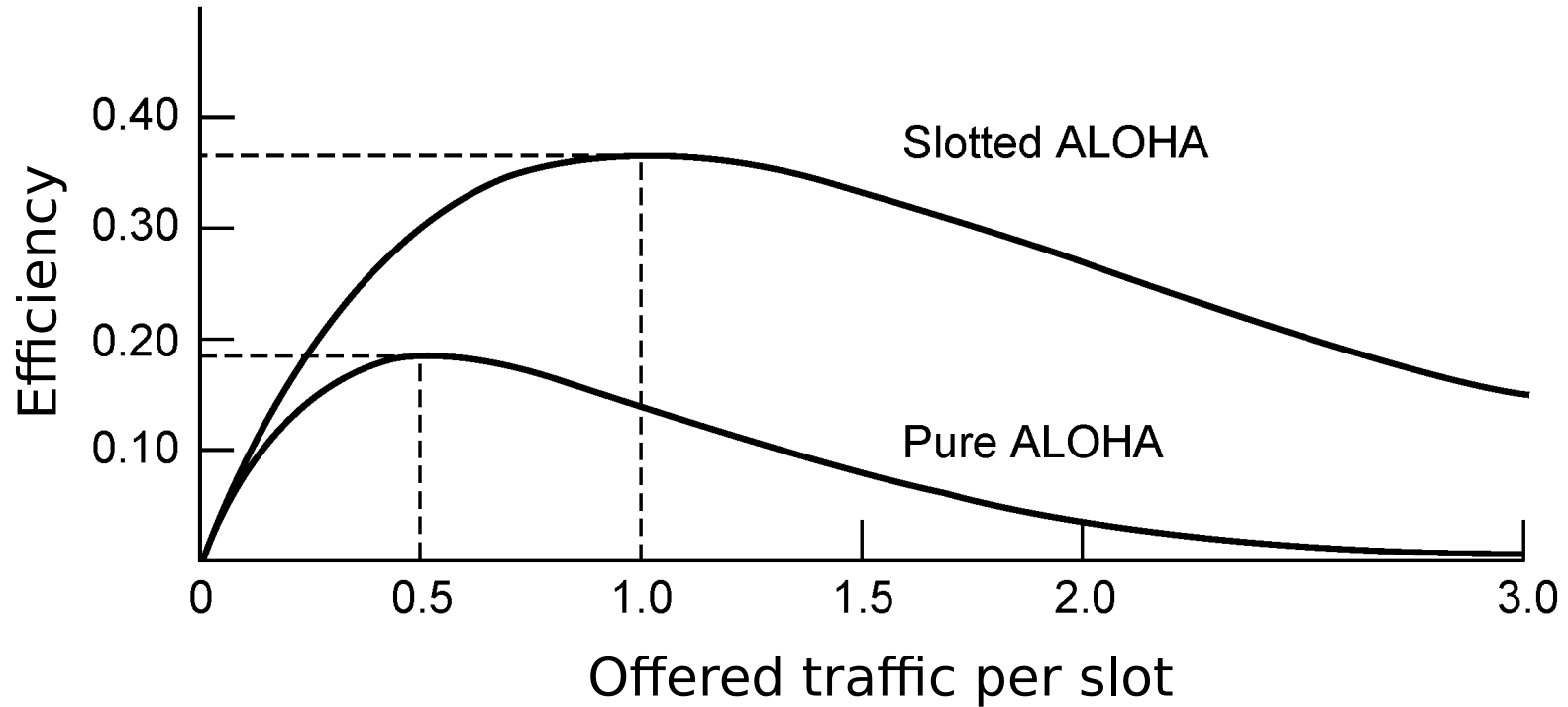
$$\text{Prob that any node has a success} = Np(1-p)^{2(N-1)}$$

Pure ALOHA efficiency

- Max efficiency: find p^* that maximizes $Np(1-p)^{2(N-1)}$

- Max efficiency = $1/2e = .18$
 - Performs even worse than slotted Aloha!
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Pure ALOHA vs Slotted ALOHA



Summary

□ Multiple Access Control:

- Random access MAC protocols
 - Slotted ALOHA, Pure ALOHA
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